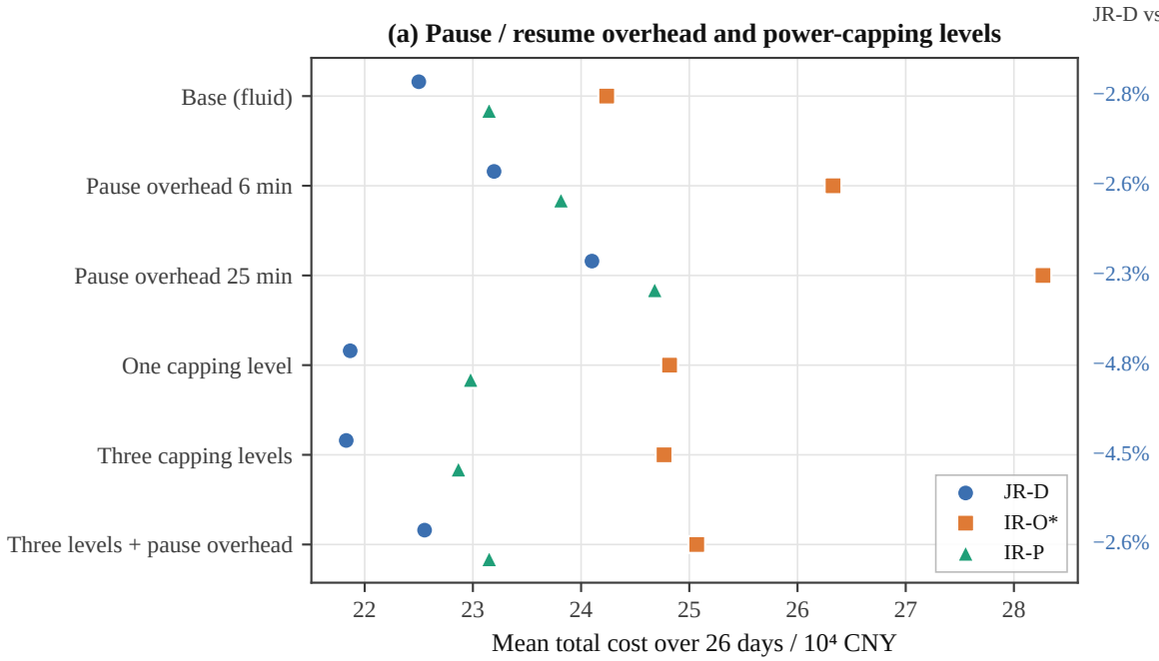
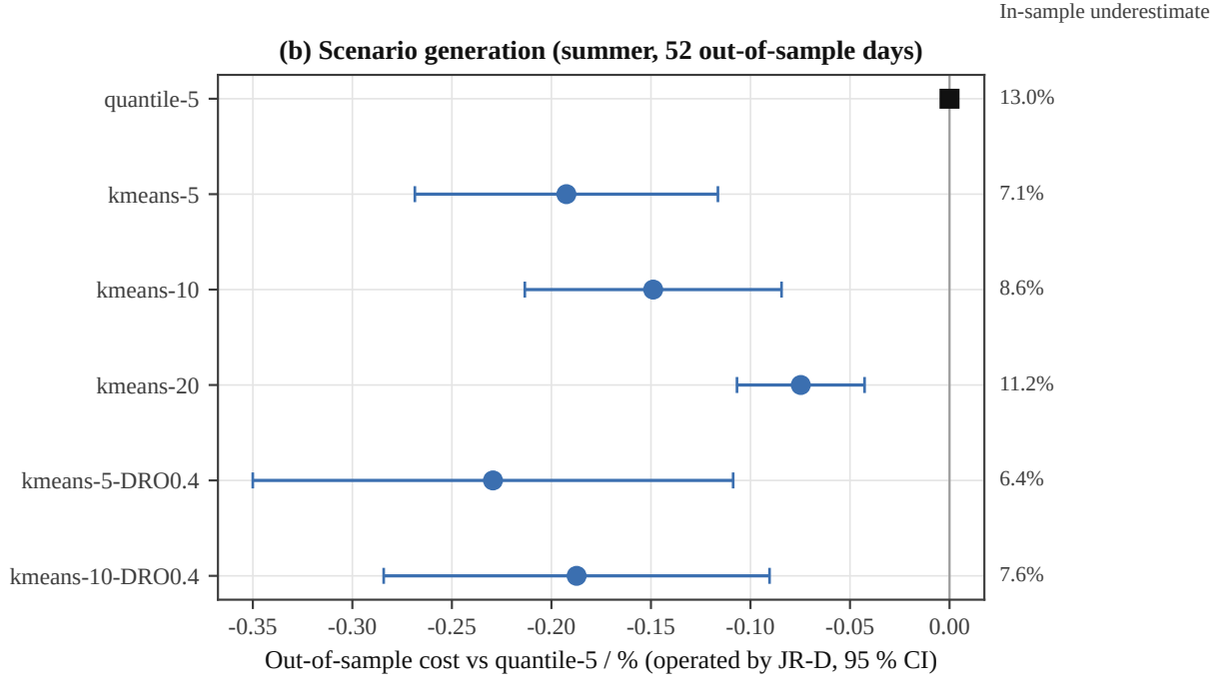
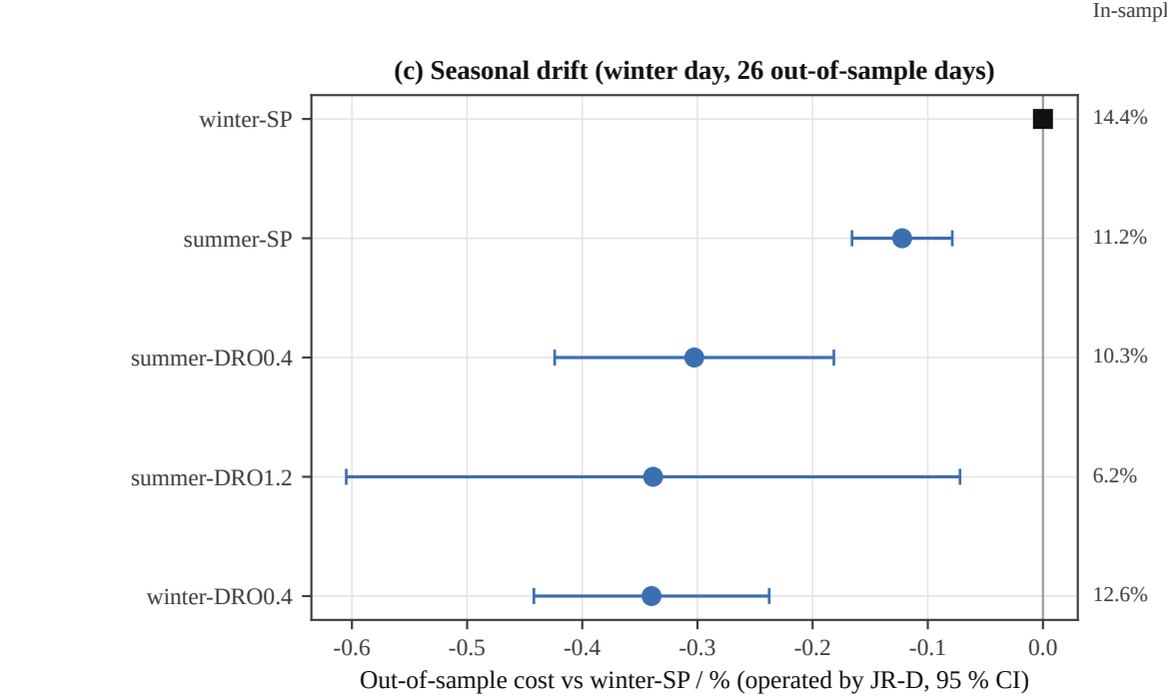
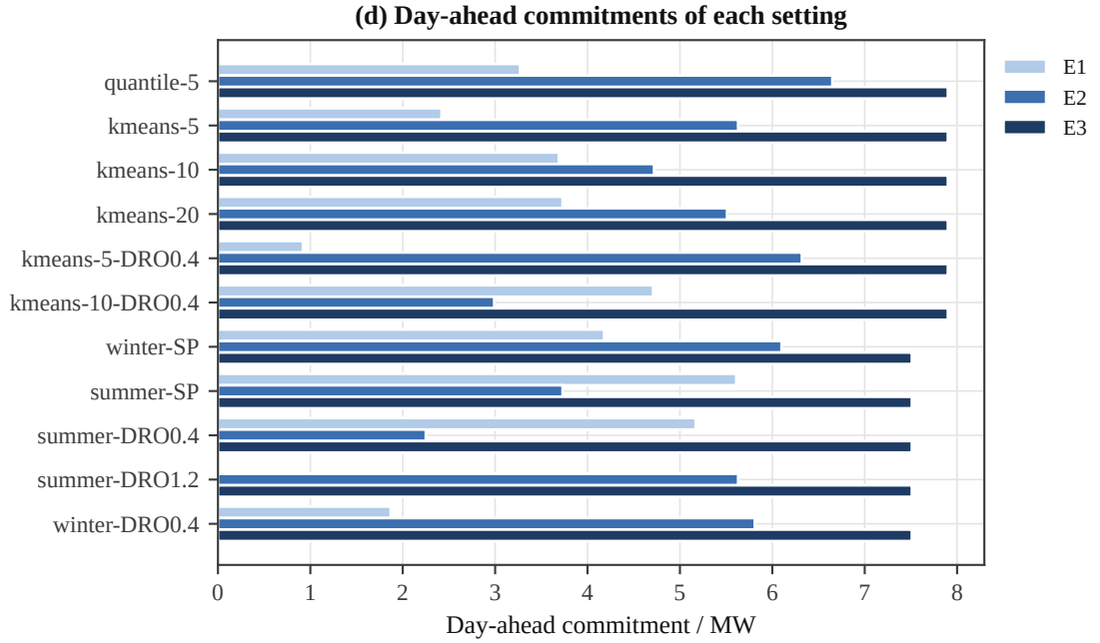

JR-D vs best IR
(a) Pause / resume overhead and power-capping levels
Base (fluid)
Pause overhead 6 min
Pause overhead 25 min
One capping level
Three capping levels
Three levels + pause overhead
−2.8%
−2.6%
−2.3%
−4.8%
−4.5%
−2.6%
JR-D
IR-O*
IR-P
22
23
24
25
26
27
28
Mean total cost over 26 days / 10⁴ CNY


In-sample underestimate
(b) Scenario generation (summer, 52 out-of-sample days)
quantile-5
kmeans-5
kmeans-10
kmeans-20
kmeans-5-DRO0.4
kmeans-10-DRO0.4
13.0%
7.1%
8.6%
11.2%
6.4%
7.6%
-0.35
-0.30
-0.25
-0.20
-0.15
-0.10
-0.05
0.00
Out-of-sample cost vs quantile-5 / % (operated by JR-D, 95 % CI)


In-sample underestimate
(c) Seasonal drift (winter day, 26 out-of-sample days)
winter-SP
summer-SP
summer-DRO0.4
summer-DRO1.2
winter-DRO0.4
14.4%
11.2%
10.3%
6.2%
12.6%
-0.6
-0.5
-0.4
-0.3
-0.2
-0.1
0.0
Out-of-sample cost vs winter-SP / % (operated by JR-D, 95 % CI)


(d) Day-ahead commitments of each setting
quantile-5
kmeans-5
kmeans-10
kmeans-20
kmeans-5-DRO0.4
kmeans-10-DRO0.4
winter-SP
summer-SP
summer-DRO0.4
summer-DRO1.2
winter-DRO0.4
E1
E2
E3
0
1
2
3
4
5
6
7
8
Day-ahead commitment / MW